AIATS for Regular Medical-2026(XII Passed)_Practice Test-1

Time: 3 Hours **Maximum Marks: 720** **Total Questions: 180**

SYLLABUS

Physics: Units & Measurements, Motion In A Straight Line, Motion In A Plane.
Chemistry: Some Basic Concepts Of Chemistry, Structure Of Atom.
Botany: Cell: The Unit Of Life, Cell Cycle And Cell Division, The Living World.
Zoology: Structural Organisation In Animals–animal Tissues, Biomolecules.

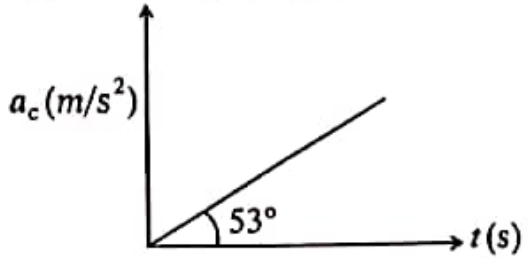
GENERAL INSTRUCTIONS:

- This question paper contains 180 multiple choice questions (MCQs).
- Each question carries 4 marks for correct answer and -1 mark for wrong answer.
- Use only Black/Blue Ball Point Pen for marking answers in OMR sheet.
- Each question has four alternatives. Choose the most appropriate alternative.
- Rough work should be done only in the space provided for rough work in the test booklet.

PHYSICS

Q.1
A particle projected from origin moves in $x - y$ plane with a velocity $\vec{v} = \hat{i} + 2x\hat{j}$, where $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y-axes. The path followed by the particle is
(1) Straight line
(2) Hyperbolic
(3) Circle
(4) Parabolic

Q.2
Centripetal acceleration (a_c) of a particle moving in a circle of radius 24 m varies with time t as shown in the figure. If particle starts from rest, then speed of the particle after 2 s will be



(1) 4 m/s
(2) 8 m/s
(3) 12 m/s
(4) 6 m/s

Q.3
Match List-I with List-II.

	List-I (Physical quantity)		List-II (Dimensions)
(A)	Electric potential (V)	(P)	$M^1L^2T^{-3}A^{-2}$
(B)	Stefan's constant (σ)	(Q)	$M^{-1}L^{-2}T^4A^2$
(C)	Capacitance (C)	(R)	$ML^2T^{-3}A^{-2}$
(D)	Resistivity (ρ)	(S)	$M^1L^2T^{-3}A^{-1}$

(1) A(P); B(Q); C(R); D(S)
(2) A(S); B(P); C(R); D(Q)
(3) A(S); B(P); C(Q); D(R)
(4) A(S); B(R); C(P); D(Q)

Q.4
If $L_1 = (1.02 \pm 0.01)$ m and $L_2 = (2.14 \pm 0.02)$ m, then $L_1 + 2L_2$ is (in m)
(1) 5.30 ± 0.01
(2) 5.30 ± 0.02
(3) 5.30 ± 0.05
(4) 3.16 ± 0.05

Q.5
In a screw gauge, the number of circular scale divisions is 100 and the pitch is 1 mm. If its least count is taken as an absolute error, then the percentage error in the measurement of length whose reading comes out to be equal to 25 mm by the screw gauge will be
(1) 2%
(2) 0.04%
(3) 0.5%
(4) 0.08%

Q.6
Ship(A) is travelling due east at 20 m/s. Another ship(B) has velocity 20 m/s, 30° north of west w.r.t. first ship. The velocity of ship (B) will be (in m/s) (assume east-direction as x-axis and north as y-axis).
(1) $20 + 10\sqrt{3}\,\hat{i} + 10\hat{j}$
(2) $20\hat{i} + 10\hat{j}$
(3) $20 - 10\sqrt{3}\,\hat{i} - 10\hat{j}$
(4) $20 - 10\sqrt{3}\,\hat{i} + 10\hat{j}$

Q.7
A ball is dropped from height of 240.1 m high tower. After 3 s, a second ball is thrown straight downwards from the same point. What should be the initial velocity of second ball so that both balls hit ground simultaneously? (g = 9.8 m/s^2)
(1) 20.46 m/s
(2) 60.28 m/s
(3) 40.42 m/s
(4) 36.44 m/s

Q.8
In 50 observations, random error is x. The random error, if the number of observations are increased by 150, will be
(1) $\frac{x}{2}$
(2) $4x$
(3) $3x$
(4) $\frac{x}{3}$

Q.9
The value of 10 J energy in a system of units when 1 unit of mass is equal to 10 g, 1 unit of length is 10 cm and 1 unit of time is equal to 2 seconds, wil be
(1) 4000
(2) 40000
(3) 400000
(4) 200

Q.10
A particle is thrown vertically up with initial velocity of 50 m/s. Ratio of the distance covered by the particle in last second of its upward journey and last second of its complete journey is (g = 10 m/s^2
(1) $\frac{5}{2}$
(2) $\frac{1}{2}$
(3) $\frac{1}{4}$
(4) $\frac{1}{5}$

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Q.11
If force (F), velocity (v) and density (d) are considered as new fundamental quantities in a new system, then dimensional formula of energy will be
(1) $F^2v^{-2}d^{-1}L^2$
(2) $F^2v^{-1}d^{-1}L^2$
(3) $F^2v^{-2}d^{-1}L^2$
(4) $F^2v^{-1}d^{-1}L^2$

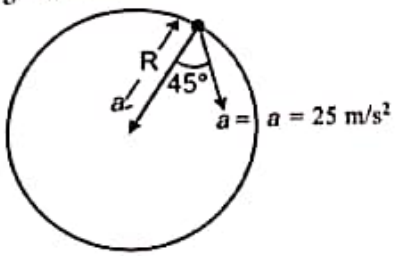
Q.12
A particle is moving along x-axis whose instantaneous speed (v) is given by, $v^2 + 2x^2 = 25$. The acceleration of the particle at $x = 2$ m is
(1) 2 m/s^2
(2) -8 m/s^2
(3) -4 m/s^2
(4) 4 m/s^2

Q.13
Number of significant figures in value of 9.99 m $-$ 0.0099 m is
(1) 5
(2) 4
(3) 3
(4) 2

Q.14
The position x of a particle depends on its velocity (v) and time (t) as given by the relation $x = Av + \frac{B}{A+t}$. Dimensions of AB is
(1) $[LT]^2$
(2) $[LT^{-2}]$
(3) $[LT^{-1}]$
(4) $[LT^{-2}]$

Q.15
A force F is applied perpendicularly on a rectangular plate of length L and breadth B. If the percentage error in the measurement of L is 2%, B is 3% and in F is 5%, what is maximum percentage error in pressure?
(1) 5%
(2) 8%
(3) 10%
(4) 1%

Q.16
If a represents the total acceleration of a particle moving in the clockwise direction in a circle of radius $R = \sqrt{2}$ m at a given instant of time as shown in the figure. The speed of the particle at given instant is



(1) $5\sqrt{2}$ m/s
(2) $\sqrt{2}$ m/s
(3) 5 m/s
(4) $2\sqrt{5}$ m/s

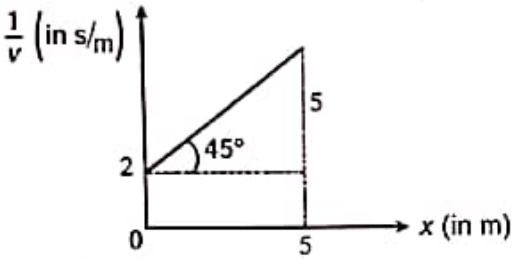
Q.17
A body projected horizontally with speed of 18 km/h from height h, takes 10 s to reach the ground. The time it takes to cover the first one-third of height will be ($g = 10$ m/s^2)
(1) 4.72 s
(2) 5.77 s
(3) 8.16 s
(4) 2.84 s

Q.18
Consider the following statements and choose the correct answer from the options given below.
Statement A: A complete set of the base units is called system of units.
Statement B: Only those physical quantities can be added or subtracted which have same dimensional formula.
Statement C: The order of magnitude of speed of light in cgs unit is 8 ×.
(1) Only statements A and C are correct
(2) Only statement B is correct
(3) Only statements B and C are correct
(4) All statements A, B and C are correct

Q.19
Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): Average velocity of a particle can be positive, negative or zero.
Reason (R): Displacement of a particle can be positive, negative or zero.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

Q.20
A boat takes 3 hours to move downstream and 5 hours to move upstream between two fixed points in a river. The time taken by boat to cover one side journey in still water is
(1) 4 hours
(2) 2.25 hours
(3) 3.25 hours
(4) 3.75 hours

Q.21
Graph of $1/v$ versus x of a particle, where v is speed (in m/s) and x is position, moving along a straight line as shown in figure. The time taken by the particle to move from $x = 0$ to $x = 5$ m is



(1) 25 s
(2) 32 s
(3) 22.5 s
(4) 18.5 s

Q.22
A body is thrown vertically upwards with speed 10 m/s, the distance covered by the body during last second of its ascent is x meter. Now another body is thrown vertically upward with speed 20 m/s, then distance covered by this body during last second of its ascent journey will be
(1) 20.46 m/s
(2) 60.28 m/s
(3) 40.42 m/s
(4) 36.44 m/s

Q.23
An open truck is moving with 72 km/h on a horizontal surface. A boy standing on the truck throws a ball upward at 30 m/s with respect to truck. The speed of the ball with respect to horizontal surface after one second will be ($g = 10$ m/s^2)
(1) 20 m/s
(2) $\frac{20}{\sqrt{7}}$ m/s
(3) $20\sqrt{2}$ m/s
(4) $\frac{10}{\sqrt{7}}$ m/s

Q.24
A displacement vector of magnitude 5 m makes an angle 30° with the x-axis. Its rectangular components in x-y plane are (in m)
(1) $\frac{5\sqrt{3}}{2}$, $\frac{5}{2}$
(2) $\frac{5\sqrt{3}}{2}$, $\frac{5}{2}$
(3) $\frac{5}{\sqrt{3}}$, $5\sqrt{2}$
(4) $5\sqrt{3}$, $\frac{5\sqrt{3}}{2}$

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Q.25

A particle is projected with velocity $\vec{u} = 2\hat{i} + 3\hat{j}$ m/s from ground. Match the physical quantities in column I with their values in appropriate SI units in column II (take $g = 10$ m/s^2)

	Column I		Column II
(A)	Time of flight	(P)	1.2
(B)	Horizontal range	(Q)	0.45
(C)	Maximum height	(R)	0.6
(D)	Landing speed	(S)	3.6

(1) A(R); B(Q); C(P); D(S)
(2) A(R); B(P); C(S); D(Q)
(3) A(Q); B(P); C(S); D(R)
(4) A(R); B(P); C(Q); D(S)

Q.26

If y represents energy and x represents pressure gradient, then the dimensions of $\frac{d^2y}{dx^2}$ are

(1) ML^5T^{-2}
(2) $M^{-1}L^6T^4$
(3) $M^{-1}L^6T^2$
(4) $M^{-1}L^{-6}T^{-2}$

Q.27

Given below are two statements : one is labelled as **Assertion** and the other is labelled as **Reason**

Assertion (A): Measured value 1.56 is more precise than 1.5.

Reason (R): More accurate measurement will always be more precise.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

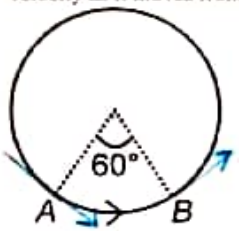
Q.28

The vector $\vec{OA}$, where O is origin, is given by $\vec{OA} = \hat{i} + \sqrt{3}\hat{j}$. Now it is rotated by 15° clockwise about O. What will be the magnitude of y-component of the new vector?

(1) 2
(2) $\sqrt{2}$
(3) $\frac{1}{\sqrt{2}}$
(4) $\sqrt{3}$

Q.29

A particle is moving on a circular path with a constant speed of 5 m/s. Magnitude of change in its velocity as it moves from A to B will be



(1) $5\sqrt{2}$ m/s
(2) 5 m/s
(3) $\frac{5}{\sqrt{2}}$ m/s
(4) $\frac{5\sqrt{3}}{2}$ m/s

Q.30

Choose the incorrect statement among the following:

(1) Momentum cannot be added to pressure
(2) Coefficient of viscosity and change in momentum are dimensionally different
(3) Rydberg constant and universal gas constant are dimensionally same
(4) Torque and kinetic energy are dimensionally same

Q.31

If L, C and R represents the physical quantities: inductance, capacitance and resistance respectively. The combination which has dimensions of frequency is

(1) $\frac{1}{RC}$
(2) LC
(3) $\frac{1}{\sqrt{LC}}$
(4) Both (1) and (3)

Q.32

Given below are two statements : one is labelled as **Assertion** and the other is labelled as **Reason**

Assertion (A): Area under velocity-time graph of an object moving in a straight line unidirectionally gives us displacement.

Reason (R): Acceleration of an object is $\vec{a} = \frac{d^2\vec{r}}{dt^2}$. (where symbols have their usual meaning)

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

Q.33

A ball is dropped from a height H. If its speed at height $\frac{H}{3}$ is 40 m/s, then height H will be

(1) 40 m
(2) 80 m
(3) 160 m
(4) 120 m

Q.34

A particle is moving along x-axis. Its position (x) varies with time (t) as $x = 4 + 2t - t^2$; where x is in meter and t is in second. At what time (t) it will reverse its motion?

(1) 2 sec
(2) 3 sec
(3) 4 sec
(4) 1 sec

Q.35

A body starts its motion from rest with constant acceleration. If the distance covered in first 2 seconds is S_1 and that covered in first 4 seconds is S_2, then

(1) $S_2 = 3S_1$
(2) $S_2 = 4S_1$
(3) $S_2 = S_1$
(4) $S_2 = 2S_1$

Q.36

A particle is moving on a circular track of radius 3 m. Its speed varies as $v = t^2$. The ratio of radial acceleration to tangential acceleration of particle at time $t = 3$ s is

(1) $\frac{2}{5}$
(2) $\frac{4}{5}$
(3) $\frac{9}{2}$
(4) $\frac{3}{4}$

Q.37

Consider the following statements and identify the correct answer:

Statement I: A projectile motion is non-uniformly accelerated motion.

Statement II: Uniform circular motion is uniformly accelerated motion.

(1) Both statement I and statement II are correct
(2) Statement I is incorrect and statement II is correct
(3) Statement I is correct but Statement II is incorrect
(4) Both statement I and statement II are incorrect

Q.38

If the displacement y of a particle is given by $y = A\sin(Pt + qx)$, then dimensional formula of AP is equal to (where x represents position and t represents time)

(1) Velocity
(2) Acceleration
(3) Area
(4) Volume

Q.39

The quantity which has same dimensions as relative density is

(1) Relative velocity
(2) Angular displacement
(3) Longitudinal strain
(4) Both (2) and (3)

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Q.40

In a village, on a rainy day a student moves to school from his home with speed 4 km/hr. He finds that school is closed due to heavy rain, so he runs back home with speed 8 km/hr. What is the average speed of student for the complete journey from home to school and then from school to home?

(1) $\frac{15}{3}$ km / hr

(2) $\frac{16}{3}$ km / hr

(3) $\frac{12}{5}$ km / hr

(4) $\frac{13}{5}$ km / hr

Q.41

In a vernier caliper, N divisions of vernier scale coincides with N–2 divisions of main scale. The least count of the instrument (in cm) is (take 1 MSD = 1 mm)

(1) $\frac{N}{10}$

(2) $\frac{1}{10\,N}$

(3) $\frac{2}{10\,N}$

(4) $\frac{2}{N}$

Q.42

A projectile that is projected from ground passes through points (2, 3) m and (3, 2) m during its motion. The range of the projectile will be

(1) $\frac{17}{8}$m

(2) $\frac{19}{5}$m

(3) $\frac{21}{4}$m

(4) $\frac{13}{5}$m

Q.43

Ship A is moving westward with a speed of 50 km h^{-1}; and another ship B is 500 km south of A, is moving northwards with a speed of 50 km h^{-1}. The minimum distance between the two ships, is

(1) $500\sqrt{2}$ km

(2) $\frac{250}{\sqrt{2}}$ km

(3) $250\sqrt{2}$ km

(4) 500 km

Q.44

A projectile is fired horizontally from the top of a high tower with speed 10 m/s. If at any time t instantaneous velocity makes an angle of 30° with the initial velocity, then t is equal to

(1) 1 s

(2) $\sqrt{3}$ s

(3) $\frac{1}{\sqrt{3}}$s

(4) $\sqrt{2}$ s

Q.45

Rain is falling vertically downwards. For a man walking at a speed 3 m/s on ground, the rain appears to fall at an angle 45° with vertical, towards the man. The speed of the rain w.r.t. ground is

(1) $3\sqrt{2}$ m/s .

(2) 3 m/s

(3) $\frac{3}{\sqrt{2}}$ m/s

(4) $2\sqrt{3}$ m/s

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Q.46

Consider the following statements:

a. Both 0.0052 and 2300 have same number of significant figures

b. Product of 2.5 × 5.25 in correct significant figures is 13.1

c. 2 books and 30 pencils have infinite number of significant figures

d. 2.008 has more number of significant figures than 2.080

The incorrect statement(s) is/are

(1) b, c and d only

(2) a and b only

(3) d only

(4) b and d only

Q.47

A hypothetical element (E) exists in 3 isotopic forms of ^{20}E, ^{21}E and ^{22}E in ratio of 5 : 2 : 1. The average atomic mass of E is

(1) 20.5 u

(2) 21.2 u

(3) 21.9 u

(4) 22.1 u

Q.48

The ratio of the total number of electrons in 3.2 g of methane to that in 3.6 g of H_2O is

(1) 1 : 1

(2) 1 : 2

(3) 2 : 1

(4) 2 : 3

Q.49

Among the following, choose the one in which maximum number of molecules are present.

(1) 10 L of H_2 gas at STP

(2) 5 g of H_2 gas

(3) 24 L of O_2 gas at STP

(4) 30 g of O_2 gas

Q.50

Consider the following compounds:

(i) Calcium carbonate

(ii) Glucose

(iii) Sulphuric acid

(iv) Carbon dioxide

The correct order of the above compounds with respect to the percentage of oxygen present in them, is

(1) (iii) > (ii) > (i) > (iv)

(2) (iv) > (iii) > (ii) > (i)

(3) (iii) > (iv) > (ii) > (i)

(4) (iv) > (ii) > (iii) > (i)

Q.51

50 g of 20% pure limestone is heated and produced carbon dioxide is passed over activated charcoal to produce carbon monoxide as follows:

$CO_2(g) + C(s) \rightarrow 2CO(g)$

The change in the gaseous volume at STP is

(1) Increased by 1.12 L

(2) Decreased by 1.12 L

(3) Increased by 2.24 L

(4) Decreased by 2.24 L

Q.52

Mass of NaOH pellets required to completely react with 500 mL of 0.01 M HCl solution according the following reaction, is

$NaOH + HCl \rightarrow NaCl + H_2O$

(1) 0.4 g

(2) 0.3 g

(3) 0.2 g

(4) 0.1 g

Q.53

2.0 g of magnesium is burnt in 0.64 g of dioxygen in a closed vessel. Which reactant is left in excess and how much?

(1) Mg and 0.96 g

(2) O_2 and 0.16 g

(3) Mg and 1.04 g

(4) O_2 and 0.24 g

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Q.54

The number of oxygen atoms produced on thermal decomposition of 2 mol of potassium chlorate ($KClO_3$) is

$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

(1) 3

(2) $3N_A$

(3) 6

(4) $6N_A$

Q.55

A mixture of 4.6 g of formic acid and 1.8 g oxalic acid is treated with conc. H_2SO_4. The evolved CO_2 and CO gases are passed through the weighed amount of KOH pellets at STP.

The increase in the mass of KOH pellets is

(Take molar mass of oxalic acid as 90 g)

(1) 0.44 g

(2) 0.88 g

(3) 1.32 g

(4) 1.54 g

Q.56

If an enzyme contains 0.01% cobalt (atomic weight = 59 u) by weight, then the minimum molecular weight of the enzyme is

(1) 5.9×10^5 u

(2) 5.9×10^4 u

(3) 2.9×10^5 u

(4) 2.9×10^4 u

Q.57

If 100 mL of 0.1 M HCl and 200 mL of 0.05 M H_2SO_4 are mixed then the molar concentration of $[H^+]$, ions in the resultant mixture will be

(1) 0.025 M

(2) 0.067 M

(3) 0.080 M

(4) 0.1 M

Q.58

Match the following columns:

	Column I (Quantity)		Column II (Number of atoms)
a.	4.48 L of carbon monoxide at STP 4	(i)	0.6 N_A
b.	3.4 g of ammonia gas 3	(ii)	0.08 N_A
c.	2.4×10^{22} molecules of hydrogen gas	(iii)	0.8 N_A
d.	8.4 g of baking soda	(iv)	0.4 N_A

(Take $N_A = 6 \times 10^{23}$)

Choose the option containing correct match:

(1) a–(iii), b–(i), c–(iv), d–(ii)

(2) a–(iii), b–(iv), c–(ii), d–(i)

(3) a–(iv), b–(iii), c–(ii), d–(i)

(4) a–(iv), b–(i), c–(ii), d–(iii)

Q.59

Consider 49% (w/w) aqueous solution of H_2SO_4 with density = 1.2 g/L and choose the incorrect statement

(1) The molality of the solution will nearly be 9.804 m

(2) Molarity of solution will be 0.06 M

(3) Mole fraction of solute will be 0.15

(4) Ratio of the moles of solute to the moles of solvent will be 5 : 28.3

Q.60

In a one molal aqueous solution of a solute, the mole fraction of solvent is approximately

(1) 0.80

(2) 0.98

(3) 0.02

(4) 0.20

Q.61

When 10 mL of 0.5 M $B(OH)_2$ (diacidic strong base) and 20 mL of 0.1 M HA (monobasic strong acid) are mixed, then the $[OH^-]$ concentration in final solution will be

(1) 0.133 M

(2) 0.152 M

(3) 0.266 M

(4) 0.367 M

Q.62

34.2 g of aluminium sulphate is dissolved in enough water to make 500 mL of solution. If aluminium sulphate dissociates completely, then the molar concentration of Al^{3+} ions and SO_4^{2-} ions respectively are

(1) 0.4 M & 0.4 M

(2) 0.2 M & 0.4 M

(3) 0.4 M & 0.6 M

(4) 0.3 M & 0.6 M

Q.63

Two metallic oxides contain 27.6% and 30% oxygen respectively if the formula of second oxide is X_2O_3 then that of first oxide will be

(1) X_3O_4

(2) XO

(3) X_3O

(4) X_2O

Q.64

SO_2Cl_2 on reaction with excess of water result into acidic mixture as follows:

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

If 8 moles of NaOH is required for the complete neutralisation of the acidic mixture then the moles of SO_2Cl_2 used is

(1) 2 mol

(2) 3 mol

(3) 4 mol

(4) 6 mol

Q.65

Consider the following statements:

(i) A given compound always contains exactly the same proportion of elements by weight

(ii) Equal volumes of gases at the same temperature and pressure should contain equal number of molecules

(iii) When gases combine or are produced in a chemical reaction, they do so in a simple ratio by volume, provided all gases are at the same temperature and pressure

Choose the option representing the correct names of the laws aligned with the given statements

	(i)	(ii)	(iii)
(1)	Avogadro law	Law of definite proportions	Gay Lussac's Law of gaseous volumes
(2)	Gay Lussac's law of Gaseous volumes	Avogadro law	Law of definite proportions
(3)	Law of definite proportions	Gay Lussac's Law of gaseous volumes	Avogadro law
(4)	Law of definite proportions	Avogadro law	Gay Lussac's law of gaseous volumes

(1) (1)

(2) (2)

(3) (3)

(4) (4)

Q.66

Haemoglobin contains 0.34% of iron by mass. Th number of Fe atoms in 5.6 g of haemoglobin is

(Atomic mass of Fe = 56 u)

(1) 2.04×10^{21}

(2) 2.04×10^{20}

(3) 1.02×10^{22}

(4) 1.02×10^{21}

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Q.67

Following are the sets of quantum numbers for 4 electrons

a. $n = 3, \ell = 1, m_\ell = +1, m_s = +\frac{1}{2}$ 3p=3px3pz✗

b. $n = 3, \ell = 2, m_\ell = -2, m_s = +\frac{1}{2}$ 3d = 5

c. $n = 4, \ell = 0, m_\ell = 0, m_s = \frac{1}{2}$ 4s = 4

d. $n = 4, \ell = 1, m_\ell = -1, m_s = \frac{1}{2}$ 4p = 5

The correct increasing order of energy of orbitals is

d > b

(1) b < c < d < a

(2) c < b < a < d

(3) a < b < c < d

(4) c < b < d

Q.68

Match the following columns I & II

	Column I (Orbital)		Column II (Number of Radial Node(s))
(a)	2p 4	(i)	3
(b)	3s 2	(ii)	2
(c)	4d 3	(iii)	1
(d)	5p	(iv)	0

Choose the correct option containing most appropriate match

(1) a – (ii), b – (iv), c – (i), d – (iii)

(2) a – (iii), b – (i), c – (iv), d – (ii)

(3) a – (iv), b – (ii), c – (i), d – (iii)

(4) a – (iv), b – (ii), c – (iii), d – (i)

Q.69

If the radius of third Bohr's orbit of hydrogen atom is r_3 and the radius of 4th Bohr's orbit of helium ion is r_4, then

(1) $r_4 = \frac{9}{8} r_3$

(2) $r_4 = \frac{8}{9} r_3$

(3) $r_4 = \frac{8}{9} r_3$

(4) $r_4 = \frac{4}{9} r_3$

Q.70

If the energy of ground state of hydrogen atom is –13.6 eV, then the energy of 1st excited state of He^+ ion in eV will be

(1) –6.04 eV

(2) –13.6 eV

(3) –19.62 eV

(4) –27.2 eV

Q.71

The number of photons of wavelength 2000 Å emitted by a bulb of power 1 watt in 2 seconds is

(1) 1.42×10^{20}

(2) 1.42×10^{19}

(3) 2.02×10^{18}

(4) 2.50×10^{20}

Q.72

Consider the following statements

a. The orbital angular momentum of an electron in s orbital is equal to zero ✓

b. The angular momentum of electron in a d orbital is equal to $\sqrt{2}\ h$

c. The value of orbital angular momentum depends on both azimuthal and spin quantum numbers ✗

Choose the option containing correct statement(s)

(1) (a) and (b) only

(2) (b) and (c) only

(3) (b) only

(4) (a) only

Q.73

The minimum percentage uncertainty in velocity of an electron which has uncertainty in position equal to its de Broglie wavelength is

(1) 5.25%

(2) 7.96%

(3) 10.20%

(4) 12.02%

Q.74

Match list I with list II

	List I (Quantum numbers)		List II (Orbital)
a.	n = 2, ℓ = 0 4	(i)	2p
b.	n = 3, ℓ = 1 3	(ii)	3d
c.	n = 3, ℓ = 2 2	(iii)	3p
d.	n = 2, ℓ = 1 1	(iv)	2s

Choose the correct answer from the options given below:

(1) a–(iii), b–(iv), c–(ii), d–(i)

(2) a–(ii), b–(iii), c–(iv), d–(i)

(3) a–(iv), b–(i), c–(ii), d–(iii)

(4) a–(iv), b–(iii), c–(ii), d–(i)

Q.75

Two electrons occupying the same orbital are distinguished by

(1) Principal quantum number

(2) Azimuthal quantum number

(3) Magnetic quantum number

(4) Spin quantum number

Q.76

Consider the following statements:

Statement I: Both d_{z^2} and $d_{x^2-y^2}$ orbitals have electron density along the axis.

Statement II: For a given subshell, there are $(2\ell + 2)$ values of m_ℓ which are possible. ✗

In the light of above statements, choose the correct option.

(1) Both the statement I and statement II are correct

(2) Both the statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

Q.77

Given below are two statements

Assertion (A): Energy of 2s orbital of hydrogen atom is greater than that of 2s orbital of lithium.

Reason (R): Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff}).

In the light of above statements, choose the correct option.

(1) Both (A) and (R) true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

Q.78

The wavelength of the light that must be absorbed to excite the electron of a hydrogen atom from level n = 2 to level n = 3 will be

(1) 656 nm

(2) 328 nm

(3) 65.6 nm

(4) 32.8 nm

$\frac{1}{\lambda} = 109677\left(\frac{1}{2^2} - \frac{1}{3^2}\right) cm^{-1}$

Q.79

Choose the incorrect statement about electromagnetic waves

(1) The oscillating electric and magnetic fields produced by oscillating charged particles are parallel to each other and both are perpendicular to the direction of propagation of wave

(2) Electromagnetic waves do not require medium to propagate

(3) There are many types of electromagnetic radiations which differ from one another in wavelength

(4) The wavelength of X-rays is less than that of UV rays

Q.80

If the work function of Lithium metal and copper metal are 2.4 eV and 4.8 eV respectively, then the ratio of maximum kinetic energies of the photon emitted by copper metal to lithium metal on providing 10 eV energy to both of them is

(1) 13 : 19

(2) 19 : 13

(3) 1 : 21

(4) 21 : 1

$\frac{KE_{Cu}}{KE_{Li}} = \frac{10 - 4.8}{10 - 2.4} = \frac{5.2}{7.6}$

Q.81

The orbital diagram in which Aufbau's principle is violated

(1) 4s 3d ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑

(2) 4s 3d ↑ ↑ ↑ ↑ ↑ ↑

(3) 4s 3d ↑↓ ↑↓ ↑↓ ↑↓ ↑ ↑

(4) 4s 3d ↑↓ ↑↓ ↑ ↑ ↓

$3d^6\ 7.2 = \frac{7.2}{\ }$ .s

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Q.82

For an electronic state 'x' of He^+ ion, Energy $E_n = -3.4$ eV

Azimuthal quantum number $\ell = 2$

Magnetic quantum number = 0

The correct statement is

(1) Total number of nodes = 4

(2) Radial node = 1

(3) It is 3d state

(4) Energy of 3s state of He^+ ion is equal to the energy of state 'x' of He^+ ion

Q.83

Choose the incorrect statement

(1) The electronic configuration of Cu is $[Ar]3d^{10}4s^1$

(2) The magnetic quantum number can have negative value

(3) Total number of unpaired electrons in Cr atom and Fe atom are same

(4) Total number of electrons in d subshell of a nickel atom is 8

Q.84

If an electron is moving with the speed of light, then the de Broglie wavelength associated with it is

(1) 1.3×10^{-13} m

(2) 2.4×10^{-12} m

(3) 2.4×10^{-11} m

(4) 1.3×10^{-14} m

Q.85

Which of the following sets of quantum numbers is not permissible?

(1) $n = 5, \ell = 3, m = +1, s = -\frac{1}{2}$

(2) $n = 4, \ell = 2, m = -2, s = +\frac{1}{2}$

(3) $n = 3, \ell = 0, m = 0, s = -\frac{1}{2}$

(4) $n = 2, \ell = 2, m = -1, s = -\frac{1}{2}$

Q.86

Match the column I with column II.

	Column I		Column II
a.	Number of orbitals in a shell	(i)	$(4\ell + 2)$
b.	Maximum number of electrons in a shell	(ii)	$(2\ell + 1)$
c.	Number of orbitals in a subshell	(iii)	$2n^2$
d.	Maximum number of electrons in a subshell	(iv)	n^2

Choose the option containing correct match

(1) a–(ii), b–(i), c–(iv), d–(iii)

(2) a–(iii), b–(iv), c–(ii), d–(i)

(3) a–(iv), b–(iii), c–(ii), d–(i)

(4) a–(iv), b–(i), c–(ii), d–(iii)

Q.87

Which of the given transitions in the hydrogen spectrum will give the photon of highest energy?

(1) n = 4 to n = 3

(2) n = 2 to n = 1

(3) n = 5 to n = 1

(4) n = 5 to n = 3

Q.88

Total number of spectral lines in H spectrum obtained in visible region when an electron jumps from n = 6 to n = 1 is

(1) 2

(2) 3

(3) 4

(4) 5

Q.89

All of the following spectral series are found in infrared region **except**

(1) Lyman

(2) Paschen

(3) Pfund

(4) Brackett

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Q.90

Which of the following have equal number of atoms?

(i) 2.4 g of H_2

(ii) 32 g of NaOH

(iii) 44 g of CO_2

(iv) 20 g of $CaCO_3$

Choose the correct set

(1) (i) & (ii)

(2) (i) & (iii)

(3) (ii) & (iv)

(4) (i), (ii), (iii) & (iv)

$\frac{2.4}{2} \times 2$ 2.4NA

$\frac{32}{40} \times 3 \times NA$

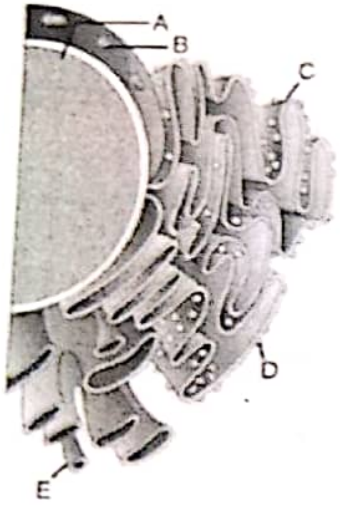
$\frac{44}{44} \times 3NA$

$\frac{20}{100} \times 5 \times NA$.

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Q.91

Observe the following figure



Based on the above given figure, certain statements are given below regarding the labels.

(a) A is absent in some mature cell *e.g.* erythrocytes of many mammals and sieve tube cells of vascular plant.

(b) B represents the structure that act as a passage through which movement of only genetic material takes place across the membrane of the organelle discovered by Robert Brown

(c) D is frequently involved in protein synthesis and secretion

(d) Lipid-like steroidal hormones are synthesised in E.

Select the **correct** option

(1) Only b and c are correct

(2) Only a and d are correct

(3) a, b and c are correct

(4) All are correct except b

Q.92

P is a non-living rigid structure which is present in young plant cell which gradually diminishes as the cell matures and Q is formed on the R side of the cell.

Select the correct option for blanks **P, Q** and **R**

	P	Q	R
(1)	Secondary cell wall	Primary cell wall	Inner (towards plasma membrane)
(2)	Primary cell wall	Secondary cell wall	Outer (towards extracellular space)
(3)	Primary cell wall	Secondary cell wall	Inner (towards plasma membrane)
(4)	Secondary cell wall	Primary cell wall	Inner (towards primary wall)

(1) (1)

(2) (2)

(3) (3)

(4) (4)

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Q.93

Select the **correct** option w.r.t. semi-autonomous organelle which is the site of oxidative phosphorylation and is not easily visible under the microscope without vital stain.

(1) Typically, it is sausage shaped or cylindrical having a diameter of 0.2 – 1.2 nm and length of 0.1 – 0.4 nm

(2) The inner membrane forms a number of infoldings called the cristae towards the cytoplasm

(3) They are found in all plant cells but not in animal cells.

(4) The inner compartment is filled with dense homogenous substance called matrix.

Q.94

Human cheek cell when observed under the microscope, the delimiting structure of the cell was different from onion cell. The delimiting structure of cheek cell is composed of

(1) Cellulose

(2) Chitin

(3) Lipids and protein

(4) Peptidoglycan

Q.95

The chromosome that appears I-shaped during anaphase

(1) Appears to have a single arm.

(2) Is also known as isobrachial chromosome.

(3) Has centromere located slightly away from the centre of chromosome.

(4) Has one extremely short and one very long arm.

Q.96

Fungal cell wall usually lacks

(a) Chitin

(b) N–acetyl glucosamine (NAG)

(c) Cellulose

(d) Peptidoglycan

The correct ones are

(1) Both (b) and (c)

(2) Both (a) and (b)

(3) Both (c) and (d)

(4) All except (a)

Q.97

The 40S smaller sub unit of 80S ribosome has

(1) 18S rRNA

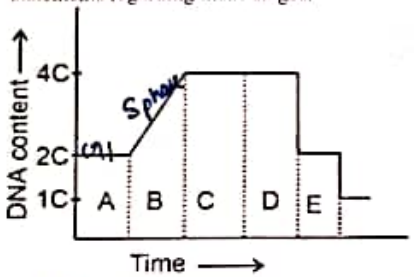
(2) 16S rRNA

(3) 23S rRNA

(4) 28S rRNA

Q.98

Given below is a schematic break-up of the phases of cell division of a diploid cell having 2C genetic content. Identify the stages and consider the statements regarding those stages.



(a) A – Tubulin protein is synthesized during this phase.

(b) E – In this phase, the nuclear membrane and nucleolus reappear in cell and four haploid nuclei are formed after the completion of this phase

(c) D – The cell after completion of this phase has half the number of chromosomes as compared to parent cell

(d) B – In plant cell, during this phase DNA replication begins in the nucleus and centriole duplicates in cytoplasm

(e) C–Semi autonomous organelles along with Golgi duplicate in this phase

Choose the **incorrect** one(s) w.r.t. cell cycle.

(1) Only (b) and (c)

(2) Only (c) and (d)

(3) All except (d)

(4) Only (a) and (d)

Q.99

Stage which shows transition to metaphase-I is also associated with

(1) Terminalisation of chiasmata

(2) Beginning of condensation of chromosomal material

(3) Recombination nodule formation

(4) Bivalent formation

Q.100

Consider the following five statements (A – D) related to cell cycle, and select the **correct** option stating them as True [T] or False [F].

(A) Nucleolus, ER, chromosomes, Golgi complex, nuclear envelope are reformed during telophase stage.

(B) The number of chromosomes doubles during S-phase.

(C) Events of cell cycle are independent of genetic control.

(D) In the S and G_2 phases, the new DNA molecules formed are not distinct but intertwined

(E) Acidic conditions are maintained inside the lysosome by pumping of H^+ ions into them.

(1) A-(T), B-(T), C-(F), D-(F), E-(F)

(2) A-(F), B-(F), C-(F), D-(T), E-(T)

(3) A-(T), B-(F), C-(T), D-(F), E-(T)

(4) A-(F), B-(F), C-(T), D-(T), E-(T)

Q.101

All of the following are the components of mitotic apparatus, **except**

(1) Asters

(2) Centrosome

(3) Chromosomes

(4) Spindle fibre

Q.102

The daughter cells which are formed after the meiosis II are genetically

(1) Similar to each other but dissimilar to parent cell.

(2) Similar to the parent cell but dissimilar to each other.

(3) Dissimilar to each other as well as to the parent cell.

(4) Similar to parent cell as well as with each other

Q.103

Match the column I with column II.

	Column I		Column II
(a)	Primata	(i)	Class
(b)	Angiospermae	(ii)	Division
(c)	Hominidae	(iii)	Order
(d)	Monocotyledonae	(iv)	Family

Choose the **correct** answer from the options given below.

(1) a–(iv), b–(iii), c–(ii), d–(i)

(2) a–(iii), b–(ii), c–(iv), d–(i)

(3) a–(i), b–(ii), c–(iii), d–(iv)

(4) a–(iii), b–(ii), c–(i), d–(iv)

Q.104

Which of the following options shows taxa with maximum number of common characters?

(1) Potato, brinjal, grasses

(2) Petunia, datura, lily

(3) Lion, tiger and leopard

(4) Tomato, lily, pea

Q.105

Who developed the binomial nomenclature for giving the scientific naming to variety of organisms?

(1) Ernst Mayr

(2) G.N. Ramachandran

(3) Robert Brown

(4) Carolus Linnaeus

Q.106

Find **odd** one w.r.t. defining features of life forms

(1) Intrinsic growth

(2) Metabolism

(3) Consciousness

(4) Acellular organisation

Q.107

The lowest category in taxonomic hierarchy is

(1) Order

(2) Genus

(3) Class

(4) Species

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State the following statements as **True (T)** or **False (F)**

(A) Schleiden proposed the hypothesis that the bodies of animals and plants are composed of cells and product of cells.

(B) Mycoplasmas are only 3.0 µm in length.

(C) In human beings, the membrane of erythrocyte has approximately 52 percent protein and 40 percent lipid.

(D) Microtubules help in the formation of cleavage furrow during cell division.

Choose the correct option

(1) A-(F), B-(T), C-(F), D-(T)
(2) A-(T), B-(T), C-(T), D-(T)
(3) A-(F), B-(F), C-(T), D-(F)
(4) A-(T), B-(T), C-(T), D-(F)

Q.109

Read the following statements and select the correct option.

Statement A: Plant, animal and fungus cannot be differentiated on the basis of the presence or absence of cell wall but can be differentiated on the basis of the composition of cell wall.

Statement B: During protein synthesis in bacteria, many ribosomes forms a chain on a rRNA and form polyribosomes.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

Q.110

Select the **correct** match.

(1) Ribosomes – They are associated with plasma membrane in prokaryotes
(2) Gram positive bacteria – Presence of membrane bound cell organelle.
(3) Gas vacuole – Mostly found in green algae.
(4) Cilia and flagella – Hair like outgrowth of cell wall

Q.111

How many events among the following occur(s) more than two times in a single meiotic division?

(a) RNA and protein synthesis
(b) DNA replication
(c) Centriole duplication

The **correct** one(s) is/are

(1) a only
(2) c only
(3) a and b only
(4) b and c only

Q.112

Choose the characteristic feature of the phase in which spindle fibres attach to kinetochores of chromosomes.

(1) Chromosomes appear like a ball of wool.
(2) The recombination between non-sister chromatids of homologous chromosomes gets completed.
(3) Nucleolus, Golgi complex and ER reform.
(4) Congression of chromosomes can be observed

Q.113

Arrange the following events in the correct chronological order and select the **correct** option.

(a) Homologous chromosomes separate.
(b) Appearance of chiasmata.
(c) Centrosome begins to move towards opposite poles of the cell.
(d) Bivalent chromosomes align themselves on equatorial plate.

(1) c → d → a → b
(2) c → b → d → a
(3) a → b → c → d
(4) b → d → c → a

Q.114

Choose the **incorrect** statements w.r.t. cell cycle and mark the correct option

(a) A cell shows growth in terms of increase in cytoplasmic content.
(b) G_0 phase of cell cycle represents a phase where cell remains metabolically active but do not proliferate.
(c) 123 mitotic divisions are required to produce 124 cells from a single cell.
(d) The first two stages of Prophase-I are relatively long lived compared to dictyotene stage.

(1) b only
(2) a, b and c
(3) d only
(4) a and c

Q.115

Given below are two statements:

Statement A: During intrameiotic interphase, chromosomes are elongated and form chromatin fibres.

Statement B: During metaphase-I, centrosome of the two chromosomes are arranged in two rows at the equator, so double metaphasic plate is formed.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both statement A and B are true.
(2) Both statement A and B are false.
(3) Statement A is true but statement B is false.
(4) Statement A is false but statement B is true.

Q.116

Match List I with List II

	List-I		List-II
(a)	Leptotene	(i)	Dissolution of synaptonemal complex
(b)	Zygotene	(ii)	Appearance of recombination nodule
(c)	Pachytene	(iii)	Beginning of compaction of chromatin material
(d)	Diplotene	(iv)	Formation of synaptonemal complex

Choose the **correct** answer from the option given below:

(1) a–(iii), b–(iv), c–(i), d–(ii)
(2) a–(iii), b–(i), c–(ii), d–(iv)
(3) a–(ii), b–(iii), c–(iv), d–(i)
(4) a–(iii), b–(iv), c–(ii), d–(i)

Q.117

Read the following statements carefully.

(a) Endoplasmic reticulum contains tubules, cisternae and vesicles.
(b) In plasma membrane, the integral proteins which run throughout the lipid bilayer are known as tunnel proteins.
(c) In *Amoeba*, the contractile vacuole is important for engulfing the food particle.
(d) Plant cell wall is chiefly composed of the water soluble polysaccharides, cellulose.
(e) In humans, a few chromosomes may have additional constrictions termed as nonstaining secondary constrictions near their ends.

Choose the **correct** option.

(1) c, d and e are correct
(2) a, b and e are correct
(3) c, d, and e are incorrect
(4) a, b and c are correct

Q.118

How many microtubules are associated with the structure of centriole and axoneme of a eukaryotic flagellum respectively?

(1) 18 and 27
(2) 27 and 20
(3) 27 and 18
(4) 18 and 20

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Q.119
In bacteria, which of the following components protect the cell and also help in adhesion?
(1) Plasmid
(2) Nucleoid
(3) Glycocalyx
(4) Pili

Q.120
All are correct w.r.t. Pseudovacuole, **except**
(1) Membraneless vacuole
(2) Found in prokaryotes
(3) Provide buoyancy
(4) Presence of tonoplast

Q.121
Which evolutionary biologist is famously known as the "Darwin of the 20th century"?
(1) Ernst Mayr
(2) Antonie Von Leeuwenhoek
(3) Rudolf Virchow
(4) G.N Ramachandran

Q.122
State the following statements as True (T) or False (F)
(a) Seven obligate taxonomic categories together constitute a taxonomic hierarchy. T
(b) Families are characterised on the basis of only reproductive features of plant species. F
(c) The science which deals with diversity of organisms and all their comparative and evolutionary relationships amongst them is systematics. T
Choose the **correct** option.
(1) a-(T), b-(F), c-(T)
(2) a-(T), b-(F), c-(F)
(3) a-(F), b-(T), c-(T)
(4) a-(F), b-(F), c-(T)

Q.123
"Growth and reproduction are mutually inclusive events". For which of the following organisms, this statement is not true?
(1) *Amoeba*
(2) *E.coli*
(3) *Euglena*
(4) *Cuscuta*

Q.124
In which of the following pairs, there is a greater difficulty of determining relationship of one taxa to the other at the same taxonomic category?
(1) Hominidae and Muscidae
(2) Anacardiaceae and Poaceae
(3) *Mangifera* and *Triticum*
(4) Mammalia and Insecta

Q.125
Read the given statements of Assertion(A) and Reason(R) and choose the **correct** option.
Assertion (A): In metaphase, chromosomes are spread through the cytoplasm of the cell.
Reason (R): Complete disintegration of the nuclear envelope marks the start of the first phase of mitosis.
(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

Q.126
The most dramatic period of cell cycle does **not** involve
(1) Splitting of the centromere
(2) Histone protein synthesis
(3) Decondensation of chromosome
(4) Disintegration of nuclear envelope

Q.127
Read the given statements of **Assertion (A)** and **Reason (R)** and choose the correct option.
Assertion (A): In prophase, chromosomal material becomes untangled during the process of chromatin condensation.
Reason (R): During prophase, each chromosome appears double and consists of two coiled sister chromatids, joined by a centromere.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

Q.128 Lys
Y is a component of endomembrane system and arise by the process of packaging in the Golgi apparatus. It is represented by simple tiny spherical sac-like structures evenly distributed in the cytoplasm
Based on the above information for Y, select the **correct** option w.r.t. 'Y'.
(1) They are rich in almost all type of hydrolytic enzymes – Lipases, Proteases, Ligase, Carbohydrases etc, active at basic pH
(2) It is specialised in the synthesis of lipids and steroids.
(3) It gives rise to SER.
(4) They are bound by a single membrane.

Q.129
Select the **incorrect** match.

	Taxon	Category
(1)	Canidae	Family
(2)	Mangifera	Genus
(3)	Poaceae	Order
(4)	pardus	Species

(1) (1)
(2) (2)
(3) (3)
(4) (4)

Q.130
Select the **incorrect** statement w.r.t. taxonomic hierarchy.
(1) Each taxonomical category in the hierarchy is referred to as a unit of classification
(2) As we go higher from species to kingdom, the number of common characters goes on decreasing.
(3) Taxonomic categories in the hierarchy are distinct biological entities.
(4) Wheat, monocot, and plants represent taxa at the same level in the taxonomical hierarchy

Q.131
Division of centromere takes place during
(1) Metaphase I
(2) Anaphase II
(3) Telophase II
(4) Metaphase II

Q.132
The word ___a___ is derived from ___b___ word ___c___ which means systematic arrangement of organisms.
Choose the option to **correctly** fill in the blanks a, b and c respectively
(1) Taxonomy, Greek and taxon
(2) Characterisation, Latin and nomos
(3) Systematics, Latin and systema
(4) Systematics, Greek and systema

Q.133
Read the following statements and select the **correct** option.
Statement A: In cytokinesis, phragmoplast grow centripetally to form cell plate.
Statement B: Meiosis involves two sequential cycles of nuclear division as well as DNA replication
(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

Q.134
A cell at G_1 phase contains 30 picogram DNA a 20 chromosomes. What will be the number of chromosomes and DNA content respectively aft G_2 phase?
(1) 20, 60
(2) 40, 90
(3) 15, 80
(4) 60, 20

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Q.135

Select the answer which gives the correct identification of the stage with its characteristics.



(1) Anaphase – Arms of the daughter chromosomes remain directed towards the pole at the leading edge and the centromere of chromosomes trail behind.

(2) Telophase – It is the best stage to study shape of chromosomes.

(3) Anaphase – Shortening of spindle fibres that are attached to the kinetochore

(4) Prophase – Centromere split and chromatids separate.

ZOOLOGY

Q.136

Biologists describe protein structure at_____levels.

Choose the correct option to fill in the blank.

(1) Three

(2) Four

(3) Two

(4) Five

Q.137

Adhering junctions

(1) Help to stop substances from leaking across a tissue

(2) Perform cementing to keep the neighbouring cells together

(3) Facilitate the cells to communicate with each other

(4) Form connection between the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules, *etc.*

Q.138

The most abundant and widely distributed tissue in the body of complex animals is named 'X'. 'X' is called so because, they

(1) Provide lining for some parts of the body

(2) Link and support other tissues/organs of the body

(3) Possess fibres that provide strength, elasticity, flexibility and excitability

(4) Perform the actions which move the body to adjust to the changes in the environment

Q.139

While performing chemical analysis of a piece of liver, it is grinded in which of the following compounds, using a mortar and a pestle to obtain a thick slurry?

(1) Cl_3CCOOH

(2) Cl_2CCH_3COOH

(3) $Cl_2CHCOOH$

(4) Cl_3COH

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Q.140

The most abundant protein in the animal world is

(1) RuBisCO

(2) Collagen

(3) Cellulose

(4) Chitin

Q.141

The epithelium which is made of a single thin layer of flattened cells with irregular boundaries is found in all of the following, **except**

(1) Intestine

(2) Air sacs of lungs

(3) Walls of blood capillaries

(4) Bowman's capsule

Q.142

Select the correct statement.

(1) Each skeletal muscle is made of many long, cylindrical fibres arranged in parallel arrays.

(2) Myofibrils are composed of numerous fine fibrils called muscle fibres.

(3) The homopolymeric protein fibres present in the connective tissue, provide strength, elasticity and flexibility to the tissue.

(4) Fibres and adipocytes are compactly packed in the matrix of adipose tissue.

Q.143

How many of the components mentioned in the box below are found in the acid-soluble fraction obtained upon chemical analysis of a living tissue?

SO_4^{2-}, PO_4^{3-}, Inulin, Glycine, GLUT-4, Palmitic acid, Glutamic acid

Select the correct option.

(1) Seven

(2) Four

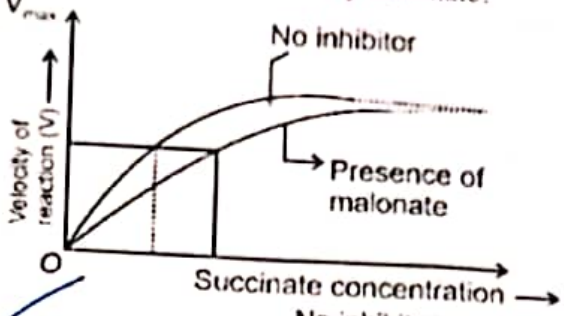
(3) Six

(4) Two

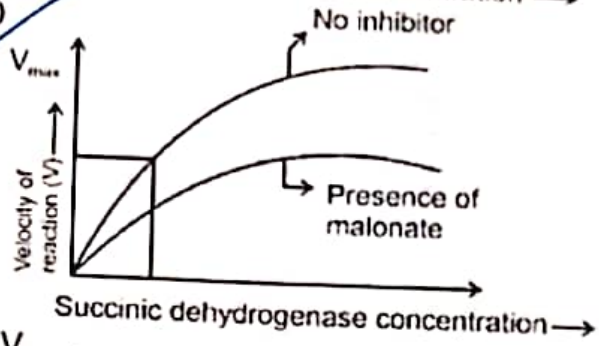
Q.144

Which of the following graphs of enzyme catalyzed reaction correctly depicts the attainment of original V_{max} when we increase the substrate concentration during competitive inhibition by malonate?

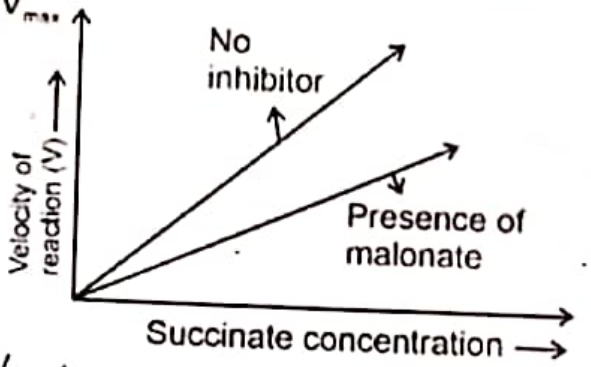
(1)



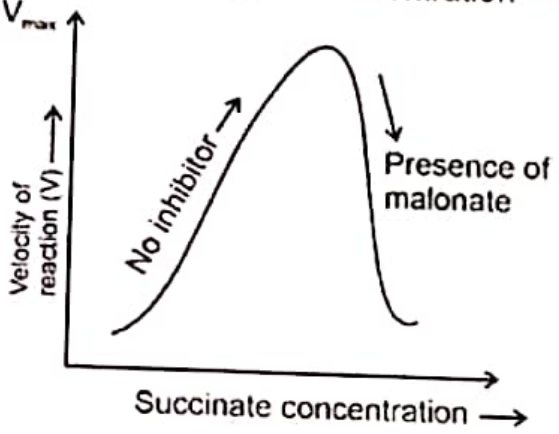
(2)



(3)



(4)



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Q.145

The epithelial cells that get specialised for secretion are

(a) Columnar cells
(b) Cuboidal cells
(c) Squamous cells

Choose the correct option.

(1) Only (b) and (c)
(2) Only (c)
(3) Only (a) and (b)
(4) Only (b)

Q.146

Consider the following cells:

(a) Macrophage
(b) Fibroblast
(c) Mast cell
(d) Adipocytes
(e) Chondrocytes
(f) Osteocytes

How many of the above given cells are found in both adipose and areolar tissue?

(1) Five
(2) Two
(3) Six
(4) Four

Q.147

Dense regular and dense irregular connective tissues are similar in all of the following aspects, **except**

(1) Orientation of fibres
(2) Presence of fibroblasts
(3) Presence of modified polysaccharides in between cells and fibres
(4) Presence of collagen

Q.148

Select the incorrect statement.

(1) When ice melts into water, there is a change in state of the matter and it represents a physical process.
(2) Hydrolysis of starch into glucose is an organic chemical reaction.
(3) Every enzyme has a primary structure, *i.e.*, linear amino acid sequence of the protein.
(4) Rates of physical processes are influenced by temperature among other factors

Q.149

In all of the following compounds, four valency positions of α-carbon are occupied by four different substituent groups, **except**

(1) Glycine
(2) Alanine
(3) Serine
(4) Valine

Q.150

Read the following statements

Statement A: Strychnine is produced by plants for their defence action.

Statement B: N-acetyl galactosamine is the nitrogen containing chemically modified sugar found in the exoskeleton of insects.

Choose the correct option.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

Q.151

The muscles which help in the movement of food through the digestive tract are made up of muscle fibres that are

(1) Branched and involuntary
(2) Cylindrical and uninucleated
(3) Fusiform in shape and multinucleated
(4) Uninucleated and unbranched

Q.152

Consider the given reaction

$$CO_2 + H_2O \overset{X}{\rightleftharpoons} H_2CO_3$$

The enzyme 'X' belongs to the class

(1) Dehydrogenases
(2) Ligases
(3) Lyases
(4) Hydrolases

Q.153

Read the following statements

(a) Stability is related to the energy status of the molecule or the structure.
(b) The energy level difference between the substrate and the product in a reaction represents whether it is a spontaneous (exothermic) or energy requiring reaction.
(c) The difference in average energy content of the substrate from that of the product is called activation energy.

Select the option representing the correct statement(s) only.

(1) (a) and (b)
(2) (b) and (c)
(3) Only (c)
(4) Only (b)

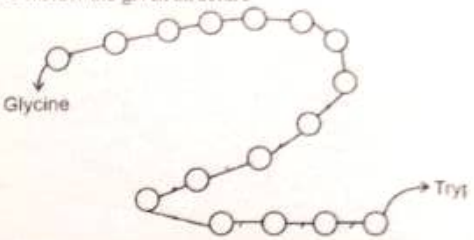
Q.154

Biomolecules which constitute 2% of the total cellular mass, is obtained in the acid-insoluble fraction upon chemical analysis of a living tissue as

(1) They are strictly macromolecules
(2) They are polymers of triglycerides
(3) When cell membranes break into pieces, water-insoluble vesicles are formed which gets separated along with the acid-insoluble part
(4) They have molecular weight more than 10,000 Da.

Q.155

Consider the given structure



Choose the correct option w.r.t the above given polypeptide.

(1) During its complete hydrolysis, 16 peptide bonds will be broken
(2) Class IV enzymes can remove the amino acid located on the left terminal of the chain
(3) The amino acid located on the N-terminal of the chain is an acidic amino acid
(4) The amino acid located on the C-terminal of the chain possesses aromatic ring in its structure.

Q.156

Read the following statements:

Statement A: All the connective tissues possess matrix, cells and fibres.

Statement B: All muscle fibres possess the property of excitability and contractility.

Select the correct option.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

Q.157

Assertion (A): Limb bones such as the long bones of the legs serve weight-bearing functions.

Reason (R): Most of the cartilages in vertebrate embryos are replaced by bones in adults.

In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(2) Both (A) and (R) are false
(3) (A) is true but (R) is false
(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

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Q.158

If a long protein chain is folded upon itself like a hollow woollen ball, then 'X' structure of protein is formed. 'X' structure will be associated with all of the following, **except**

(1) Three dimensional view of the protein
(2) Necessary for biological activities in living organisms
(3) Exhibits hydrogen and disulphide bonds
(4) Provides positional information of amino acids

Q.159

The muscle fibres present in the muscles which are associated with humerus are

(1) Voluntary and fusiform shaped
(2) Branched and striated
(3) Uninucleated and branched
(4) Cylindrical and multinucleated

Q.160

Read the following statements
(a) Epithelial tissues possess tight junctions.
(b) Muscle fibres present in the wall of stomach are held together with the help of cell junctions as they are bundled together in a connective tissue sheath.
(c) Intercalated discs (which are modified adhering junctions) at all fusion points allow the cardiac myocytes to contract as a unit.
(d) In epithelial tissue, the cells are loosely packed with little intercellular matrix.
How many of the above given statements is/are true?

(1) Four
(2) Zero
(3) Two
(4) Three

Q.161

Select the odd one w.r.t components obtained upon hydrolysis of nucleotides found in RNA.

(1) Nitrogenous base
(2) Phosphoric acid
(3) Pentose sugar
(4) Hexose sugar

Q.162

In an experiment, two reactions X and Y were conducted in different conditions. In reaction X, platinum (inorganic catalyst) was used while in reaction Y, catalase was used. Now during these reactions, many differences were noted in the different catalysts used. Select the correct point of difference from the table given below.

	Conditions	Platinum	Catalase
(1)	High pH (above optimum level)	Unaffected	Unaffected
(2)	High temperature (above optimum level)	Unaffected	Affected
(3)	Substrate	Highly specific	Non-specific
(4)	High pressure	Affected	Unaffected

(1) (1)
(2) (2)
(3) (3)
(4) (4)

Q.163

The epithelium of proximal convoluted tubule of nephron has

(1) Tall and slender cells with microvilli for increasing surface area for absorption
(2) Flattened cells with irregular boundaries for easier diffusion
(3) Cube-like cells with cilia to move particles in a specific direction in the nephron
(4) Cube-like cells with microvilli to increase surface area for absorption

Q.164

Which of the following components is characteristically absent in the structure of dense regular connective tissue

(1) Collagen fibres
(2) Elastic fibres
(3) Fibroblasts
(4) Myofibrils

Q.165

Choose the correct match w.r.t humans.

(1) Cuboidal epithelium – In the tip of nose
(2) Areolar tissue – Ligament
(3) Squamous epithelium – Lining of stomach
(4) Dense irregular connective tissue – Present in the skin

Q.166

The phenomenon where a monomeric compound exists in dipolar ionized form under certain pH condition is the characteristic of
(a) Glutamic acid
(b) Alanine
(c) Cysteine
(d) Chitin
Choose the correct option.

(1) (a) and (c) only
(2) (c) and (d) only
(3) (a), (b) and (c) only
(4) (a), (b), (c) and (d)

Q.167

The unicellular exocrine gland distributed along the epithelial lining of the GIT is known as

(1) Pancreas
(2) Goblet cells
(3) Salivary glands
(4) Sweat glands

Q.168

The epithelial lining of the bronchioles predominantly has
(a) Tall and slender cells
(b) Motile cilia at the apical surface
(c) Cells with basal nuclei
Select the correct option.

(1) (b) and (c) only
(2) (a) and (b) only
(3) (b) only
(4) (a), (b) and (c)

Q.169

Enzymes that catalyse removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds belong to the class of enzymes known as

(1) Hydrolases
(2) Lyases
(3) Transferases
(4) Dehydrogenases

Q.170

Select the incorrect statement.

(1) In the absence of any enzyme, the chemical reactions in living organisms will be very slow.
(2) Catalytic activity is lost when the co-factor is removed from the enzyme
(3) Carbonic anhydrase markedly accelerates the reaction of carbonic acid into carbon dioxide and water by 1 million times relative to the uncatalyzed reaction.
(4) There are vast number of catalytically active proteins in living organisms, each uniquely catalysing particular biochemical transformations.

Q.171

Assertion (A): The terminal residue at the left end of the glycogen corresponds to the reducing end.
Reason (R): Free functional group, such as carboxyl group, present at the left end of the glycogen participates in the redox reactions.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) (A) is true but (R) is false
(3) Both (A) and (R) are false
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Q.172

In humans, the type of epithelium that is composed of multiple layers of cells is located in

(1) Fallopian tubes
(2) Tubular parts of nephron
(3) Inner lining of pancreatic duct
(4) Walls of blood vessels

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Q.173

How many of the components mentioned in the box below is/are present in the monosaccharides which polymerise to form starch?

Hydroxyl, Carboxylic, Aldehyde, Phosphate, Carbonyl, Amino

Select the correct option.

(1) Five
(2) Four
(3) Two
(4) Three

Q.174

The excitable neural cells make up

(1) One-half of the volume of connective tissue in our body.
(2) More than one-half the volume of neural tissue in our body.
(3) Less than one-half of the volume of neural tissue in our body
(4) More than one-third of the volume of connective tissue in our body.

Q.175

Select the odd one w.r.t polymeric substances.

(1) Gum
(2) Rubber
(3) Cellulose
(4) Morphine

Q.176

All of the following entities represent a compound comprising of nitrogenous base which is covalently linked to a pentose sugar, **except**

(1) Adenosine
(2) Guanosine
(3) Cytosine
(4) Thymidine

Q.177

In humans, epithelium lining the moist surface of the buccal cavity

(1) Is made up of single layer of flattened cells
(2) Performs only exclusive role of secretion of mucus.
(3) Provides protection against chemical stresses.
(4) Secretes digestive enzymes from the endocrine glands

Q.178

Select the correct option to complete the analogy w.r.t. co-factors.

Peroxidase : Haem : : Carboxypeptidase :_____

(1) Haem
(2) Zinc
(3) Niacin
(4) NAD

Q.179

Match the column I with column II.

	Column I		Column II
(a)	Inulin 2	(i)	Primary metabolite
(b)	Cellulose 3	(ii)	Homopolymer
(c)	Uridine 1	(iii)	Secondary metabolite
(d)	Cholesterol	(iv)	Heteropolymer
(e)	Trypsin 5	(v)	Homocyclic ringed structure

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv), e(v)
(2) a(ii), b(iii), c(v), d(iv), e(i)
(3) a(ii), b(iii), c(i), d(v), e(iv)
(4) a(ii), b(iii), c(i), d(iv), e(v)

Q.180

Bones differ from cartilages as

(1) Former is a semi-rigid structure rich in calcium salts
(2) Latter is composed of chondrocytes that are also present within bone marrow
(3) Former is composed of collagen fibres, calcium salts and osteocytes
(4) Latter is composed of collagen fibres and chondrocytes that are found in lamellae

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